



AIC, MATHEMATICS LEARNING AREA  
YEAR 11 MATHEMATICS APPLICATIONS – UNIT 1

Assessment type: Response

TASK 4 – TEST 3

Student Name: \_\_\_\_\_ ID: \_\_\_\_\_ Date: \_\_\_\_\_

**TIME ALLOWED FOR THIS PAPER**

**Reading and Working time for this paper: 50 minutes in class under test conditions**

**MATERIAL REQUIRED FOR THIS PAPER**

*TO BE PROVIDED BY THE SUPERVISOR*

Question/answer booklet for sections one and two.

*TO BE PROVIDED BY THE CANDIDATE*

*Standard Items:* pens, pencils, pencil sharpener, highlighter, eraser, ruler, drawing templates

**IMPORTANT NOTE TO CANDIDATES**

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

**Structure of this paper**

| Section                      | Number of questions available | Suggested working time (minutes) | Marks available |
|------------------------------|-------------------------------|----------------------------------|-----------------|
| <b>1- Non-calculator</b>     | <b>4</b>                      | <b>15</b>                        | <b>15</b>       |
| <b>2- Calculator assumed</b> | <b>6</b>                      | <b>35</b>                        | <b>30</b>       |
|                              |                               | <b>Marks available:</b>          | <b>45</b>       |
|                              |                               | <b>Task Weighting</b>            | <b>8%</b>       |

**Instructions to candidates**

- The rules for the conduct of this examination are detailed in the booklet *WACE Examinations Handbook*. Sitting this examination implies that you agree to abide by these rules.
- Answer the questions in the spaces provided.
- Spare answer pages can be used. If you need to use them, indicate in the original answer space where the answer is continued.

Pythagoras' theorem

- 1.3.1 use Pythagoras' theorem to solve practical problems in two dimensions and for simple applications in three dimensions

Mensuration

- 1.3.2 solve practical problems requiring the calculation of perimeters and areas of circles, sectors of circles, triangles, rectangles, parallelograms and composites
- 1.3.3 calculate the volumes of standard three-dimensional objects, such as spheres, rectangular prisms, cylinders, cones, pyramids and composites in practical situations, for example, the volume of water contained in a swimming pool
- 1.3.4 calculate the surface areas of standard three-dimensional objects, such as spheres, rectangular prisms, cylinders, cones, pyramids and composites in practical situations, for example, the surface area of a cylindrical food container

Similar figures and scale factors

- 1.3.5 review the conditions for similarity of two-dimensional figures, including similar triangles
- 1.3.6 use the scale factor for two similar figures to solve linear scaling problems
- 1.3.7 obtain measurements from scale drawings, such as maps or building plans, to solve problems
- 1.3.8 obtain a scale factor and use it to solve scaling problems involving the calculation of the areas of similar figures and surface areas and volumes of similar solids

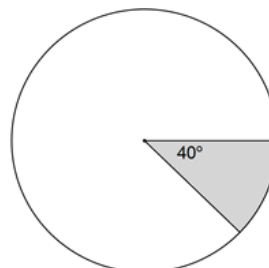
**Non-Calculator section 1:**

**Total marks:**        /15 marks

**Time allocated:** 15 minutes

**Question 1 [2 marks]**

The circular spinner drawn right has an area of  $234 \text{ cm}^2$ . Determine the area of the shaded section



**Question 2 [1, 2 = 3 marks]**

A fire started 600 m due East of Jason's house and 800 m due South of Izzy's house. Jason wants to go from his house to Izzy's house.

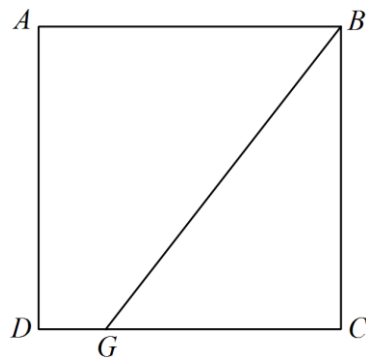
(i) Draw a labelled diagram to show the locations of the two houses and the fire.

(ii) Determine the shortest distance from Jason's house to Izzy's house.

**Question 3 [1, 1, 1, 2 = 5 marks]**

ABCD is a square with sides of 4 cm. BG divides the square into a triangle BCG and a trapezium ABGD. The distance from D to G is 1 cm.

a) Calculate the length of BG.



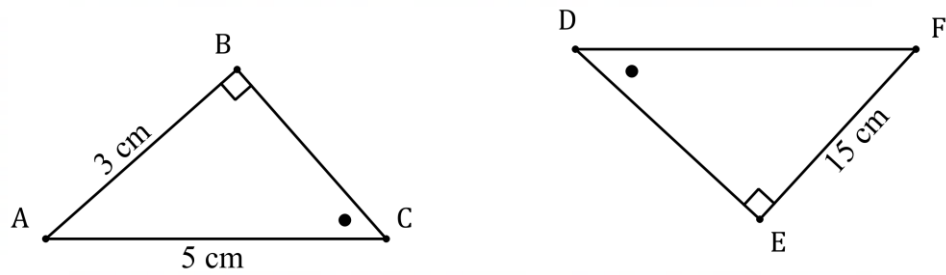
b) Calculate the area of triangle BCG.

c) Calculate the area of the trapezium ABGD.

d) Express the area of the trapezium as a percentage of the square's area.

**Question 4 [1, 3, 1 = 5 marks]**

The diagram below, not drawn to scale, shows two similar right triangles.



- Calculate the scale factor for the larger triangle relative to the smaller.
- Determine the length of side  $BC$ , the length of side  $DE$  and the length of side  $DF$ .
- Calculate how many times greater the area of the large triangle is compared to the area of the small triangle.

**Calculator Allowed section 2:**

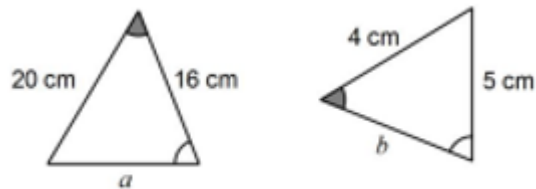
**Name:** \_\_\_\_\_

**Total marks:**        /30 marks

**Time allocated:** 35 minutes

**Question 1 [2, 2 = 4 marks]**

- (a) The two triangles shown below are similar. Determine the lengths  $a$  and  $b$ .



- (b) An image, with one side that is 13 cm long, is enlarged so that the same side now measures 39 cm. If the original area of the image was  $200 \text{ cm}^2$ , determine the area of the enlargement.

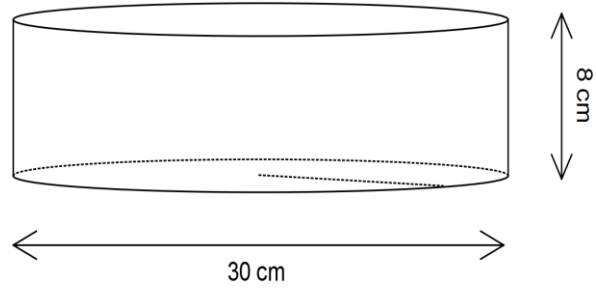
**Question 2 [2, 2 = 4 marks]**

Red Stag energy drink is sold in cylindrical cans of radius 4 cm and height 15 cm. To advertise Redd Stag the company constructs plastic blow-up scale models of the can.

- (a) The model has a radius of one metre, determine the height of the model. Give your answer in metres.
- (b) The capacity of a can is  $240\pi$  mL. State the capacity of the model in litres. Leave your answer in terms of  $\pi$ .

**Question 3 [2, 2, 2 = 6 marks]**

Fatima has baked a cake to celebrate Eid. It is cylindrical in shape, 30 cm wide and 8 cm high.



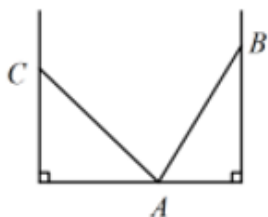
- (c) Fatima wishes to place paper trimming right around the outside of the cake. Her trimming is 8 cm wide. What is the minimum length that it should be? Give your answer to 2 decimal places.
- (b) Fatima is going to ice the cake on the top and around the outside (not on the bottom).
- i) How much of the cake will need to be iced? Give your answer to the nearest whole number.
- ii) If Fatima needs 100 g of sugar for each  $1000 \text{ cm}^2$  of icing that she makes, how much sugar will she need? Give your answer to 2 decimal places.



**Question 4 [1, 3, 2 = 6 marks]**

A 4.5 m long ladder rests on level ground between two buildings with one end, A, on the ground and the other end touching point B, at a height of 4.25 m up the wall of one of the buildings. The ladder is then rotated about A so that the other end touches point C, 3.6 m up the opposite wall.

- (a) Add the above dimensions to the sketch below.

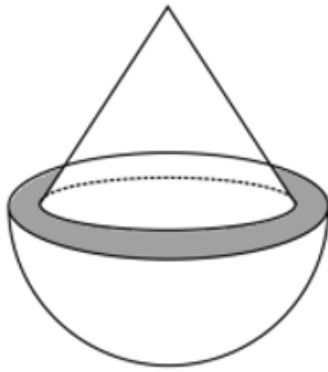


- (b) Determine the horizontal distance between the two buildings, giving your answer correct to two decimal places

- (c) A length of wire is stretched tightly from B to C. Determine the length of this piece of wire.

**Question 5 [1, 2, 3 = 6 marks]**

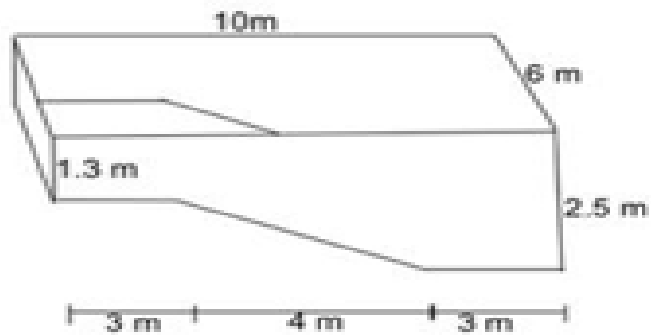
A solid cone of radius 9 cm and height 12 cm is placed symmetrically atop a solid hemisphere of radius 11 cm to form the composite solid shown below.



- (a) Use Pythagoras' Theorem to calculate the slant height of the cone.
- (b) Determine the area of the shaded ring, between the cone and the hemisphere, as shown in the diagram above. Give your answer to two decimal places.
- (c) Determine the surface area of the composite solid. Give your answer to two decimal places.

**Question 6 [2, 2 = 4 marks]**

The diagram gives the internal dimensions of a swimming pool.



- (a) The pool is to be filled with water to a level that is 15 cm below the top edge of the pool. Find the volume of water needed to fill the pool in kL.
- (b) How long will it take to fill this pool to the required level if water to this pool can be supplied at a constant rate of 120 litres per minute? Give your answer in hours and minutes.

END OF TEST

## **EXTRA WORKING OUT PAGE**